

INWOO HWANG

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SUMMARY

AI Researcher and Ph.D. Candidate at Seoul National University specializing in **Generative Motion Foundation Models**, **Embodied AI**, and **3D/4D Computer Vision**. Research focuses on building controllable and physically grounded motion intelligence, spanning **kinematic motion generation/reconstruction** to **physics-based embodied control**. Expertise includes multi-scale autoregressive and controllable diffusion-based human motion generation, online egocentric motion reconstruction, human-world interaction modeling, and reinforcement learning for humanoid control.

WORK EXPERIENCE

Snap Research

Research Scientist Intern (Generative Human-Scene Modeling)

New York City, NY

05/2024 – 09/2024

- Pioneered **4D human-scene interaction reconstruction** from **real-world monocular videos**, capturing complex physical interaction dynamics for AR and spatial computing applications.
- Led novel diffusion framework modeling human-scene interactions in real-world environments, achieving an **ICCV 2025 Highlight**.

EDUCATION

Seoul National University

Ph.D. Candidate in Electrical and Computer Engineering (Advisor: Prof. Young Min Kim)

Seoul, Korea

03/2022 – Current

Seoul National University

B.S. in Electrical and Computer Engineering

* Mandatory military service completed (12/2017 – 12/2019).

Seoul, Korea

03/2016 – 02/2022

Seoul Science High School for the Gifted

Seoul, Korea

03/2013 – 02/2016

RESEARCH EXPERIENCE

Generative Motion Foundation Models

- **Multi-Scale Autoregressive Modeling.** Developed *ScaleMoGen* (ECCV'26), a **multi-scale** skeletal-temporal token mapping pipeline driven by **autoregressive next-scale prediction**; established new **state-of-the-art** benchmarks in **text-to-motion** generation and unlocked zero-shot motion editing.
- **Masked Sequence Modeling.** Contributing to *SnapMoGen* (MoMask++) (NeurIPS'25), developing a highly scalable **masked modeling** paradigm that synthesizes intricate, high-fidelity human behaviors from expressive language descriptions.

Robust and Online Motion Control & Human-World Interaction Modeling

- **Online Reconstruction.** Architected *EgoForce* (UnderReview), a stable long-horizon motion reconstruction framework tailored for **streaming egocentric observations**, with causal **diffusion forcing** for real-time **online inference**.
- **Noisy Robust Interaction Reconstruction.** Developed *SceneMI* (ICCV'25, Highlight), a novel diffusion-based **human-scene modeling** pipeline that robustly recovers physically grounded interaction dynamics from **noisy real-world observations** obtained from IMU sensors or monocular videos.
- **Intuitive Sparse & Flexible Control.** Developed a **hierarchical diffusion** framework (CVPRW'25) for **intuitive** goal-driven motion generation from scene geometry. This work extends *SFControl* (ICCV'25), a **flexible timeline-independent** formulation and is capable of mapping highly **sparse keyjoint signals** into high-fidelity motion.

Physics-Based Control & Embodied RL

- **Physics-Grounded Adaptation.** Leading an ongoing project to build physics-based motion authoring layers that dynamically adjust trajectories with centroidal dynamics optimization, enabling data-efficient humanoid control across diverse environments from minimal reference data in simulation (*Isaac Gym*).

3D Modeling & Differentiable Neural Rendering

- **3D Stylization.** Developed *Text2Scene* (CVPR'23, Highlight) for understanding part information from text labels and scene-level **texture synthesis** via **differentiable mesh rendering**, and *Text2PointCloud* (Eurographics'23) for stylizing point clouds.
- **Implicit Scene Reconstruction.** Formulated *Ev-NeRF* (WACV'23), an **implicit neural rendering** paradigm to reconstruct high-fidelity 3D representations from asynchronous streaming sensory inputs under extreme noise.

PUBLICATIONS

• [1] EgoForce: Robust Online Egocentric Motion Reconstruction via Diffusion Forcing Inwoo Hwang, Donggeun Lim, Hojun Jang, and Young Min Kim	Under Review [Paper] [Project]
• [2] ScaleMoGen: Autoregressive Next-Scale Prediction for Human Motion Generation Inwoo Hwang, Hojun Jang, Bing Zhou, Jian Wang, Young Min Kim, and Chuan Guo	ECCV 2026 [Paper] [Project]
• [3] Accelerated Likelihood Maximization for Diffusion-based Versatile Generation Hyunsoo Lee, Inwoo Hwang, and Young Min Kim	ECCV 2026 [Paper] [Project]
• [4] SnapMoGen: Human Motion Generation from Expressive Texts Chuan Guo, Inwoo Hwang, Jian Wang, and Bing Zhou	NeurIPS 2025 [Paper] [Project]
• [5] SceneMI: Motion In-betweening for Modeling Human-Scene Interaction Inwoo Hwang, Bing Zhou, Young Min Kim, Jian Wang, and Chuan Guo	ICCV 2025 Highlight [Paper] [Project]
• [6] SFCtrl: Motion Synthesis with Sparse and Flexible Keyjoint Control Inwoo Hwang, Jinseok Bae, Donggeun Lim, and Young Min Kim	ICCV 2025 [Paper] [Project]
• [7] Less is More: Improving Motion Diffusion Models with Sparse Keyframes Jinseok Bae, Inwoo Hwang, Youngyoon Lee, Ziyu Guo, Joseph Liu, Yizhak Ben-Shabat, Young Min Kim, Mubbasir Kapadia	ICCV 2025 [Paper]
• [8] Event-Driven Storytelling with Multiple Lifelike Humans in a 3D Scene Donggeun Lim, Jinseok Bae, Inwoo Hwang, Seungmin Lee, Hwanhee Lee, and Young Min Kim	ICCV 2025 [Paper] [Project]
• [9] A Survey on Human Interaction Motion Generation Kewei Sui, Inwoo Hwang*, Anindita Ghosh*, Bing Zhou, Jian Wang, and Chuan Guo	IJCV 2026 [Paper] [Project]
• [10] Goal-Driven Human Motion Synthesis in Diverse Tasks Inwoo Hwang, Jinseok Bae, Donggeun Lim, and Young Min Kim	CVPRW 2025 [Paper] [Project]
• [11] Versatile Physics-based Character Control with Hybrid Latent Representation Jinseok Bae, Jungdam Won, Donggeun Lim, Inwoo Hwang, and Young Min Kim	Eurographics 2025 [Paper] [Project]
• [12] Text2Scene: Text-driven indoor scene stylization with part-aware details Inwoo Hwang, Hyeonwoo Kim, and Young Min Kim	CVPR 2023 Highlight [Paper] [Project]
• [13] Text2PointCloud: Text-Driven Stylization for Sparse PointCloud Inwoo Hwang, Hyeonwoo Kim, Donggeun Lim, Inbum Park, and Young Min Kim	Eurographics 2023 [Paper] [Project]
• [14] Ev-NeRF: Event Based Neural Radiance Field Inwoo Hwang, Junho Kim, and Young Min Kim	WACV 2023 [Paper] [Project]
• [15] Ev-TTA: Test-Time Adaptation for Event-Based Object Recognition Junho Kim, Inwoo Hwang, and Young Min Kim	CVPR 2022 [Paper] [Project]
• [16] MasKGrasp: Mask-based Grasping for Scenes with Multiple General Real-world Objects Junho Lee, Junhwa Hur, Inwoo Hwang, and Young Min Kim	IROS 2022 [Paper] [Project]

HONORS AND AWARDS

• Doctoral Consortium • European Conference on Computer Vision (ECCV)	2026
• Doctoral Consortium • Korean Conference on Computer Vision (KCCV)	2026
• Young Researcher Award • SNU Institute of New Media and Communications (INMC)	2026
• Excellent Research Talent Fellowship (\$20,000) • Brain Korea 21 (BK21)	2023, 2025
• The Hyundai Motor Ph.D. Fellowship (\$40,000) • Chung Mong-Koo Foundation	2022 – Present
• The Presidential Science Scholarship (Mathematics Major) (\$40,000) • Ministry of Science and ICT	2016 – 2022
• Gold Medal • National University Mathematics Competition (For Mathematics Majors)	2020

TECHNICAL SKILLS

Generative Modeling: • Diffusion Models (Diffusion Transformers (DiT), Conditional/Controllable Sampling, Online Diffusion, Diffusion Forcing) • Autoregressive Models (Next-Scale Token Map Prediction, Multi-Scale Token Maps, Masked Sequence Modeling) • Text-to-Motion • Controllable Synthesis.

Physics Simulation & Embodied RL: Isaac Gym, Reinforcement Learning, Imitation Learning, Physics-based Animation Pipelines.

3D/4D Computer Vision: Video-based 4D (Human & Scene) Reconstruction, Neural Radiance Fields (NeRF), Differentiable Rendering.

Infrastructure & Core Tools: PyTorch, JAX, Distributed Large-scale Training, CUDA, Ubuntu/Linux, macOS, Git, C/C++, Python, Unreal Engine, Blender.

ACADEMIC SERVICES

Reviewer: CVPR (2023–2026), ICCV (2023, 2025), ECCV (2024, 2026), NeurIPS (2025, 2026), TOG (2026), SIGGRAPH / SIGGRAPH-Asia (2026), ICRA (2023), RA-L (2023–2026), 3DV (2025–2026), BMVC (2026).

TEACHING EXPERIENCE

• Linear Algebra for Electrical Systems (Graduate Teaching Assistant)	2025 Fall
• Seminar in Electrical and Computer Engineering (Graduate Teaching Assistant)	2026 Spring